

```
#include <bits/stdc++.h>

using namespace std;

const int MxN = 200020;

vector<int> adj[MxN];
set<int> color[MxN];
int sz[MxN];

void merge(set<int> &a, set<int> &b) {
    if(a.size() < b.size()) {
        swap(a, b);
    }
    for(auto e: b) {
        a.emplace(e);
    }
}

void dfs(int u, int p) {
    for(auto v: adj[u]) {
        if(v == p) {
            continue;
        }
        dfs(v, u);
        merge(color[u], color[v]);
    }
    sz[u] = color[u].size();
}

signed main(int argc, char *argv[]) {
    cin.tie(nullptr)->ios::sync_with_stdio(false);
    int n;
    cin >> n;
    for(int i=1, c; i<=n; ++i) {
        cin >> c;
        color[i].emplace(c);
    }
    for(int i=2, u, v; i<=n; ++i) {
        cin >> u >> v;
        adj[u].emplace_back(v);
        adj[v].emplace_back(u);
    }
    dfs(1, 0);
    for(int i=1; i<=n; ++i) {
        cout << sz[i] << " ";
    }
    cout << "\n";
    return 0;
}
```